

Supplementary Material

Witnesses Explain Anomalies

Anonymous Submission

This supplement collects extended tables, figures, and a second case study. Full proofs of all results are in the main paper. Theorem, proposition, equation, and section numbers refer to the main paper. The main paper proves Lemma 1, Theorem 1 and Theorem 3 in place; the two longer proofs are collected here.

I. DEFERRED PROOFS

Proof of Proposition 1 (Extreme-value baseline)

Fix $u \in \mathbb{S}^{d-1}$ and let $z_i = u^\top x_i$. By Assumption 1 the variables $z_i - \mathbb{E}[z_i]$ are i.i.d. sub-Gaussian with proxy variance bounded by σ^2 (projection onto u preserves the sub-Gaussian property with the operator-norm constant $u^\top \Sigma u \leq \sigma^2$). By the maximal sub-Gaussian inequality [1], $\mathbb{E}[\max_i (z_i - \mathbb{E}[z_i])] \leq \sigma\sqrt{2\log n}$, and concentration around the expectation gives

$$\max_{i \leq n} (z_i - \mathbb{E}[z_i]) \leq \sigma\sqrt{2\log n} + \sigma\sqrt{2\log(1/\delta)}$$

with probability $1 - \delta$ [2]. Substituting $\delta = 1/n$ and adding the $\log 2/\sqrt{2\log n}$ envelope slack produces $\sigma c_d(n)$ with residual $O_P(1/\sqrt{\log n})$ (the slack is a chosen constant, not the Gumbel expansion, used only for a clean deterministic upper bound). For the location/scale rescaling, classical results [3] give $\text{med}(z) = \mathbb{E}[z_i] + O_P(n^{-1/2})$ and $\text{MAD}(z) = \sigma \Phi^{-1}(3/4) + O_P(n^{-1/2})$ under our sub-Gaussian assumption with continuous CDF; dividing the displayed bound by $\text{MAD}(z)$ absorbs the residual into $O_P(1/\sqrt{\log n})$ since $1/\sqrt{n} = o(1/\sqrt{\log n})$. Uniformity over $u \in \mathbb{S}^{d-1}$ uses an ϵ -net $\mathcal{N}_\epsilon \subset \mathbb{S}^{d-1}$ of cardinality $\leq (3/\epsilon)^d$ [4]: as $u \mapsto u^\top x_i$ is $\|x_i\|$ -Lipschitz, replacing u by its nearest net point changes $(z_i - \text{med})/\text{MAD}$ by $O_P(\epsilon\sqrt{\log n})$ uniformly in i , so $\epsilon = 1/\sqrt{n \log n}$ makes the net slack $o_P(1)$ while the union bound costs $\log |\mathcal{N}_\epsilon| = O(d \log n)$, absorbed into the $O_P(\sqrt{d/\log n})$ residual. $\square$

Proof of Theorem 2 (Plug-in consistency)

Fix any $x \in X$; the per-point score is $s_K(x) = \frac{1}{K} \sum_{k=1}^K \tau(x, u_k)$ with the $u_k \stackrel{\text{iid}}{\sim} \text{Unif}(\mathbb{S}^{d-1})$, and $s^*(x) = \mathbb{E}_u[\tau(x, u)]$. Under Assumption 1 the per-direction excess satisfies the deterministic envelope $0 \leq \tau(x, u) \leq M_n$ with $M_n = C_\sigma \sqrt{\log n}$, by combining Proposition 1 with the MAD- z definition of τ . Hoeffding's inequality [4] on the bounded summands $\tau(x, u_k) \in [0, M_n]$ gives $\mathbb{P}[|s_K(x) - s^*(x)| > t] \leq 2 \exp(-2Kt^2/M_n^2)$. Union-bounding over the n points and choosing $t = M_n \sqrt{\log(2n/\delta)/(2K)}$ yields $\sup_{x \in X} |s_K(x) - s^*(x)| \leq C_\sigma \sqrt{\log n \log(n/\delta)/K}$ with probability $1 - \delta$, as stated. $\square$

II. EXTENDED TABLES

A. Default hyper-parameters

Table I lists the single default configuration used in all experiments (no per-dataset tuning).

TABLE I
WAND DEFAULT HYPER-PARAMETERS USED IN ALL EXPERIMENTS.

Symbol	Default
K (probe budget)	1024
Spacing k (NN rank)	$\lceil \sqrt{n} \rceil$
Axis probes	on
Mix weight λ	0.25
Seed replicates S	3

B. Per-dataset detection results

Table II reports per-dataset ROC-AUC for all 16 unsupervised detectors over the 47 ADBench tasks (the main paper reports the aggregate summary).

C. Pixel-level explanation on raw images

WAND's score is differentiable in its input features, so when those features come from a differentiable encoder ϕ , $\partial(\text{WAND score})/\partial(\text{pixels})$ is obtained by autograd through ϕ , with no training. We embed the ANOCUB images with a frozen pre-trained ResNet-18, fit WAND on the 512-d embeddings (z-scored, no whitening; detection AUC 0.99), and back-propagate the anomaly score of a flagged bird to its pixels (SmoothGrad). The main paper's Fig. 1 shows the resulting saliency for three flagged anomalies and a normal inlier on a shared scale: it concentrates on the anomalies' discriminative body/bill regions and is weak on the inlier. This realises the pixel-level explanation; embedding-space detection (AUC 0.99) is on par with the named concept space (1.0), the trade-off being only that embedding dimensions are not individually named, so the map is spatial-only. Note that, as in the concept space, we fit WAND *without* Mahalanobis whitening: on the contaminated 512-d embedding covariance, whitening suppresses the anomaly directions and drops AUC to 0.88.

D. The ANOCUB dataset

ANOCUB is the explainable-AD benchmark we derive from CUB-200-2011 [5]. Each bird image is represented by its 312 named binary attributes (the class-level concept profile); inliers are the sparrow species and anomalies are a small set of birds from very different families (pelican, frigatebird,

TABLE II
ROC-AUC OVER 47 ADBENCH DATASETS FOR 16 UNSUPERVISED DETECTORS. **BOLD** = BEST PER ROW, UNDERLINE = SECOND. “-” = BUDGET / MEMORY EXHAUSTED.

Dataset	<i>n</i>	<i>d</i>	Contam.	IForest	LOF	OCSVM	KNN	PCA	HBOS	ECOD	COPOD	ABOD	COF	SOD	INNE	LODA	LSCP	KDE	PIDForest	WAND
breastw	683	9	35.0%	0.988	0.443	0.951	0.977	0.956	0.985	<u>0.991</u>	0.994	-	0.428	0.934	0.697	0.986	-	0.984	0.975	0.990
glass	214	7	4.2%	0.788	0.770	0.599	0.864	0.654	0.812	0.705	0.755	0.843	<u>0.868</u>	0.887	0.774	0.655	0.799	0.820	0.776	0.787
Hepatitis	80	19	16.2%	0.732	0.719	0.721	0.727	0.752	0.775	0.739	0.804	0.566	0.479	0.631	0.621	0.623	0.777	0.649	0.713	0.790
Ionosphere	351	32	35.9%	0.842	0.866	0.849	<u>0.928</u>	0.784	0.561	0.728	0.789	0.921	0.860	0.884	0.890	0.788	0.813	0.938	0.797	0.907
Lymphography	148	18	4.1%	0.999	0.993	0.995	0.995	0.996	0.995	0.995	0.996	0.979	0.995	0.934	0.984	0.877	0.994	0.989	0.978	<u>0.998</u>
Pima	768	8	34.9%	0.664	0.601	0.624	<u>0.709</u>	0.648	<u>0.709</u>	0.594	0.654	0.667	0.591	0.582	0.681	0.601	0.638	0.723	0.675	0.687
Stamps	340	9	9.1%	0.895	0.591	0.872	0.774	<u>0.907</u>	0.904	0.876	0.930	0.762	0.540	0.772	0.841	0.883	0.832	0.890	0.876	0.901
vertebral	240	6	12.5%	0.352	<u>0.445</u>	0.420	0.378	0.377	0.305	0.420	0.335	0.365	0.473	0.387	0.383	0.295	0.351	0.317	0.277	0.259
WBC	223	9	4.5%	0.995	0.821	0.992	0.986	<u>0.993</u>	0.988	0.995	0.995	0.935	0.758	0.977	0.933	0.990	0.974	0.973	0.992	0.995
WDBC	367	30	2.7%	0.987	0.982	0.983	0.974	0.986	<u>0.992</u>	0.971	0.994	0.885	0.947	0.947	0.975	0.980	0.990	0.950	0.991	0.983
wine	129	13	7.8%	0.795	0.879	0.696	0.519	0.806	<u>0.915</u>	0.733	0.867	0.417	0.302	0.446	0.785	0.834	0.907	0.582	0.812	0.939
WPBC	198	33	23.7%	0.486	0.520	0.485	0.500	0.482	<u>0.548</u>	0.481	0.523	0.490	0.474	0.474	0.500	0.502	0.530	0.489	0.537	0.564
annthyroid	7200	6	7.4%	<u>0.832</u>	0.727	0.681	0.811	0.673	0.620	0.789	0.776	-	0.708	0.794	0.700	0.465	0.731	0.684	0.879	0.777
cardio	1831	21	9.6%	0.926	0.546	<u>0.935</u>	0.686	0.950	0.840	<u>0.935</u>	0.922	-	0.567	0.623	0.913	0.893	0.712	0.748	0.860	0.923
Cardiotoco.	2114	21	22.0%	0.665	0.524	0.696	0.491	<u>0.752</u>	0.620	0.785	0.663	0.452	0.539	0.492	0.666	0.675	0.563	0.503	0.609	0.653
fault	1941	27	34.7%	0.567	0.596	0.539	<u>0.720</u>	-	0.572	0.469	0.455	0.699	0.563	0.661	0.573	0.495	0.575	0.731	0.571	0.486
landsat	6435	36	20.7%	0.472	0.547	0.424	0.576	0.364	0.559	0.368	0.421	0.503	0.543	<u>0.577</u>	0.541	0.380	0.560	0.625	0.456	0.575
letter	1600	32	6.2%	0.648	0.899	0.598	0.901	0.525	0.588	0.572	0.560	0.886	0.889	<u>0.909</u>	0.700	0.533	0.850	0.924	0.664	0.688
PageBlocks	5393	10	9.5%	0.904	0.716	0.915	0.834	0.905	0.760	0.914	0.875	0.740	0.624	0.662	0.947	0.719	0.813	0.907	0.854	<u>0.916</u>
pendigits	6870	16	2.3%	0.947	0.499	0.931	0.743	<u>0.936</u>	0.926	0.927	0.905	0.657	0.523	0.659	0.894	0.924	0.705	0.891	0.927	0.926
satellite	6435	36	31.6%	0.695	0.542	0.664	0.665	0.601	0.754	0.583	0.633	0.549	0.536	0.599	0.742	0.615	0.638	<u>0.760</u>	0.649	0.781
satimage-2	5803	36	1.2%	0.993	0.536	<u>0.997</u>	0.932	0.977	0.978	0.965	0.974	0.759	0.559	0.771	0.998	0.982	0.859	0.964	0.983	0.993
thyroid	3772	6	2.5%	<u>0.978</u>	0.665	0.959	0.959	0.956	0.950	0.977	0.939	-	0.587	0.876	0.971	0.816	0.788	0.958	0.967	0.980
vowels	1456	12	3.4%	0.767	0.943	0.778	0.977	0.610	0.677	0.593	0.496	<u>0.966</u>	0.960	0.914	0.898	0.635	0.921	0.956	0.740	0.885
Waveform	3443	21	2.9%	0.705	0.706	0.672	0.723	0.640	0.694	0.603	0.734	<u>0.639</u>	0.697	0.650	<u>0.748</u>	0.635	0.742	0.751	0.629	0.566
Wilt	4819	5	5.3%	0.441	<u>0.701</u>	0.317	0.613	0.434	0.349	0.394	0.345	0.634	0.723	0.595	0.359	0.403	0.598	0.334	0.507	0.427
yeast	1484	8	34.2%	0.394	<u>0.457</u>	0.420	0.406	0.396	0.399	0.444	0.381	0.416	0.449	<u>0.477</u>	0.394	0.504	0.434	0.383	0.413	0.397
ALOI	49534	27	3.0%	0.539	-	-	-	<u>0.548</u>	0.530	0.530	0.515	-	-	-	0.558	0.509	-	-	0.538	0.542
celeba	202599	39	2.2%	0.687	-	-	-	0.784	0.749	0.757	0.751	-	-	-	0.757	0.627	-	-	0.654	0.782
cover	286048	10	1.0%	0.882	-	-	-	<u>0.934</u>	0.712	0.920	0.884	-	-	-	0.959	0.833	-	-	0.933	0.855
donors	619326	10	5.9%	0.768	-	-	-	0.824	0.742	0.888	0.815	-	-	-	0.733	0.968	-	-	0.628	<u>0.909</u>
fraud	284807	29	0.2%	0.951	-	-	-	0.953	<u>0.954</u>	0.950	0.947	-	-	-	0.956	0.621	-	-	0.947	0.941
http	567498	3	0.4%	0.999	-	-	-	0.996	0.994	0.979	0.992	-	-	-	<u>0.998</u>	0.281	-	-	0.992	0.994
magic.gamma	19020	10	35.2%	0.719	0.693	-	0.811	0.667	0.712	0.638	0.681	<u>0.784</u>	0.638	0.746	0.735	0.649	-	0.722	0.744	0.681
mammography	11183	6	2.3%	0.861	0.719	-	0.838	0.894	0.830	0.906	<u>0.905</u>	-	0.717	0.805	0.824	0.891	0.741	0.865	0.843	0.881
shuttle	49097	9	7.2%	0.997	-	-	-	0.990	0.984	0.993	0.995	-	-	-	0.989	0.641	-	-	0.972	0.996
skin	245057	3	20.8%	0.667	-	-	-	0.417	0.595	0.489	0.471	-	-	-	0.722	0.432	-	-	<u>0.727</u>	0.894
smtp	95156	3	0.0%	0.904	-	-	-	0.808	0.800	0.880	0.912	-	-	-	<u>0.925</u>	0.829	-	-	0.918	0.956
backdoor	95329	193	2.4%	0.742	-	-	-	-	0.740	0.846	0.789	-	-	-	0.905	0.237	-	-	0.733	0.894
campaign	41188	62	11.3%	0.699	0.625	-	0.741	-	0.766	0.770	0.783	0.724	-	-	0.731	0.500	-	-	<u>0.781</u>	0.667
census	299285	500	6.2%	0.607	-	-	-	-	-	-	-	-	-	-	<u>0.665</u>	-	-	-	0.532	0.672
InternetAds	1966	1555	18.7%	0.701	0.609	-	0.652	0.615	<u>0.696</u>	0.677	0.676	0.585	0.614	0.562	0.610	0.523	-	0.616	0.648	0.619
mnist	7603	78	9.2%	0.809	0.673	-	0.837	<u>0.850</u>	0.576	0.746	0.774	0.755	0.635	0.675	0.858	0.426	0.643	0.798	0.621	0.849
musk	3062	166	3.2%	1.000	0.636	-	0.616	1.000	1.000	0.956	0.946	0.050	0.560	0.601	1.000	0.992	0.915	0.071	1.000	1.000
optdigits	5216	62	2.9%	0.733	0.537	-	0.395	0.515	0.869	0.605	0.682	0.513	0.511	0.512	0.615	0.608	0.569	0.323	<u>0.794</u>	0.576
SpamBase	4207	57	39.9%	0.627	0.457	-	0.489	0.550	<u>0.662</u>	0.656	0.688	0.400	0.447	0.500	0.543	0.372	0.566	0.495	0.650	0.582
speech	3686	400	1.7%	0.466	0.509	-	0.491	0.469	0.476	0.470	0.491	0.763	0.532	<u>0.570</u>	0.478	0.496	-	0.451	0.472	0.446
Avg. AUC				<u>0.762</u>	0.658	0.730	0.729	0.741	0.743	0.744	0.748	0.655	0.624	0.688	0.759	0.655	0.727	0.708	0.750	0.777
Avg. rank				<u>6.04</u>	10.11	8.80	7.54	7.81	7.40	7.79	7.23	10.29	11.57	10.14	6.71	10.13	8.65	8.69	7.43	5.64
Share of wins				5.4	0	0	2	2.2	1.2	2.2	6.2	1	2	1	<u>7.2</u>	2	0	6	1.2	7.4

mallard, hummingbird), giving a 1259×312 task with 15 anomalies. As ANOCUB is not a standard download, we release the construction script, which regenerates the exact task (anocub_task.npz) from the public CUB archive, together with the embedding/saliency scripts below.

a) *Whitening ablation.*: On the ResNet-18 embedding used for the pixel-level view (Fig. 1), Mahalanobis whitening on the contaminated covariance suppresses the anomaly directions; turning it off recovers detection (Table III), no backbone change is needed.

b) *Backbone.*: Among lightweight ImageNet backbones (z-scored embeddings, no whitening), ResNet-18 gives the best embedding-space detection (Table IV); we therefore keep it for the pixel-level explanation.

TABLE III
ANOCUB DETECTION AUC: RESNET-18 EMBEDDING, WITH/WITHOUT MAHALANOBIS WHITENING.

ResNet-18 embedding	whiten=ON	whiten=OFF
raw	0.882	0.980
z-scored	0.881	0.989

E. Explanation gallery and failure modes

Figure 1 illustrates both explanation modes on the same task and is deliberately fair about where each is weak.

a) *What is, and is not, missed.*: In the named-concept space all 15 anomalies rank in the top 15 (detection AUC 1.0), so there is no detection miss; 14/15 attribute most strongly to



Fig. 1. ANOCUB explanation gallery and failure modes. *Top*: concept-level part heatmaps (witness attribution mapped to CUB keypoints, hotter = more responsible). *Bottom*: pixel-level saliency through a frozen ResNet-18 (shared scale). Green titles = clean, red = where that mode is weak. The two modes have *complementary* blind spots. The small Anna Hummingbird the embedding nearly misses (bottom right, 90th pct: a perched hummingbird fills a sparrow-sized region and lands near the inlier manifold, so its saliency is weak and diffuse) is flagged cleanly in concept space (top middle, 99th pct, localised to the bill). Conversely the Mallard whose single most-responsible concept is the *global* “duck-like” shape, which has no body-part keypoint, so the part heatmap cannot localise it (top right), is sharply localised by pixel saliency (bottom middle).

TABLE IV
ANOCUB DETECTION AUC BY LIGHTWEIGHT BACKBONE (Z-SCORED, NO WHITENING).

backbone	dim	AUC
ResNet-18	512	0.989
EfficientNet-B0	1280	0.975
MobileNetV3-small	576	0.971

bill length, which maps to the beak keypoint and localises cleanly. The lone exception is a Mallard whose top concept is the global “duck-like” shape ($\approx 15\%$ of its attribution mass falls on global attributes with no keypoint): the part-grounded view cannot pin it to a location, even though the named reason is still correct and human-readable. In the ResNet-18 embedding space detection is near-perfect (AUC 0.99): the cross-family large birds (pelican, frigatebird, mallard) are flagged at the 99th percentile with crisp saliency, but the small hummingbirds are harder, one Anna Hummingbird drops to the 90th percentile (about 130 sparrow inliers score above it) because at 224×224 it embeds close to the sparrows. We report these openly: neither mode dominates, and the concept- and pixel-level views are complementary rather than redundant, the case each one struggles with is handled by the other.

F. Case study: named-feature medical data

On Breast Cancer Wisconsin (benign cases as inliers, a few malignant cases as anomalies; AUC 0.933, no labels used), WAND attributes flagged malignant cases to mean

radius, mean/worst perimeter, worst area, and mean concavity, the clinically recognised malignancy markers, read directly off the witness directions (Figure 2).

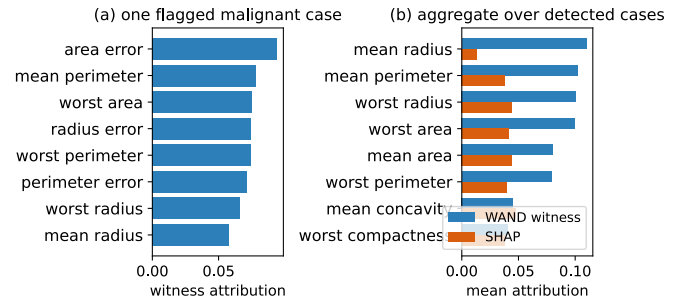


Fig. 2. Breast Cancer Wisconsin. (a) Witness attribution for one flagged malignant case; (b) aggregate over detected cases, WAND witness vs. SHAP, both concentrate on size/shape malignancy markers.

G. Critical-difference diagram (AUPR / AP)

The main paper ranks the detectors by ROC-AUC; to check that the ordering is not an artefact of that metric, Figure 3 repeats the Nemenyi post-hoc analysis on AUPR / average precision, which weighs the positive (anomaly) class more heavily. The two rankings agree.

H. AUC vs. runtime

Beyond accuracy alone, Figure 4 plots detection quality against wall-clock cost, restricted to the 22 datasets that *every*

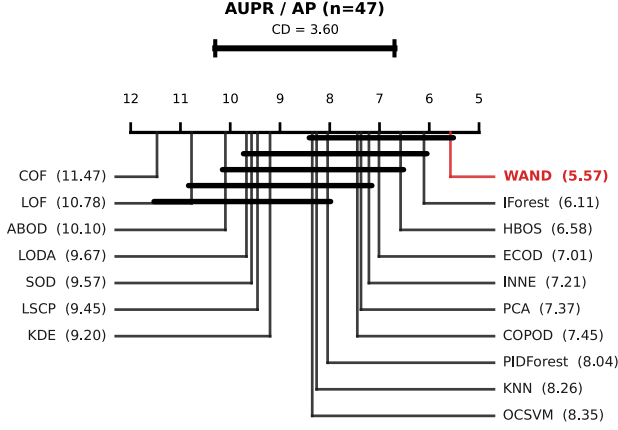


Fig. 3. Critical-difference diagram on AUPR / AP (Nemenyi post-hoc, $\alpha = 0.05$, $CD = 3.60$); the ROC-AUC diagram is Figure 3 in the main paper. The AP ranking mirrors the ROC ordering.

method completes so the per-method means cover the same tasks. WAND lands on the top-left Pareto frontier: no baseline is at once faster and more accurate on this shared subset.

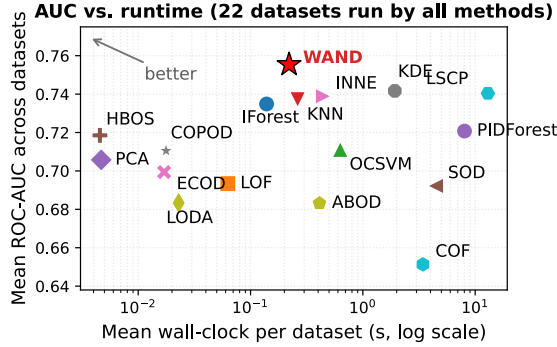


Fig. 4. Mean ROC-AUC vs. mean wall-clock per dataset (log x), averaged over the 22 ADBench datasets completed by *every* method so all means cover the same tasks. WAND sits on the AUC-runtime Pareto frontier (top-left): no other method is both faster and more accurate on this shared subset. The subset necessarily excludes the largest datasets, on which the neighbour-based methods (LOF, kNN, ABOD, COF, SOD, LSCP, KDE, OCSVM) exceed the time budget, whereas WAND runs on all 47 (Table II).

I. Scalability to large n

WAND is *inductive*: it calibrates once on a reference sample (per-direction median/MAD, weights, and the sub-Gaussian baseline), then scores arbitrary points in a single pass. The probe budget K is *independent of* n , so the deployed scorer is $O(Knd)$, linear in n , the same complexity class as Isolation Forest / ECOD / HBOS and unlike the $O(n^2)$ neighbour methods. Points are scored independently against the frozen background, so scoring streams in constant-memory chunks (bit-for-bit identical to a single call: $\max |\Delta| = 0$ on 2×10^5 points) and is embarrassingly parallel over rows. Figure 5 confirms this on synthetic data ($d=20$, an 8-core CPU): after

a 1.1 s calibration on 50k points, the default $K=1024$ scorer processes 10^7 points in 164 s ($\approx 6.1 \times 10^4$ pts/s) and the lite $K=256$ setting in 47 s ($\approx 2.1 \times 10^5$ pts/s); throughput is flat in n , so wall-clock grows strictly linearly out to $n=10^7$. The probe budget K is therefore a direct speed/quality knob, and WAND scales to high-throughput streams, the probe-efficiency guarantee is a budget, not a runtime, claim, but the runtime itself is linear and small.

J. Dataset statistics

Table V lists the 47 ADBench tasks.

III. ADDITIONAL CLARIFICATIONS

This section collects detailed answers to natural questions about cost, calibration, and scope. We are explicit about what we have measured and what we leave to future work.

a) Cost of the spacings step and memory footprint.:

For each of the K directions we (i) form the projections $z_i = u^\top x_i$ in $O(nd)$ and (ii) sort them *once*, $O(n \log n)$. The single sort yields all order statistics $z_{(1)} \leq \dots \leq z_{(n)}$, and from them every two-sided k -spacing $d_{k,i}$ in one $O(n)$ pass over ranks; the MAD branch reuses the same sorted array for the median/MAD, so no second sort is needed. The per-direction cost is therefore $O(n(d + \log n))$ and the total $O(Kn(d + \log n))$ (Remark 1); the d axis probes reuse the coordinates directly. The working set is $O(n)$ per direction (the projected/sorted vector), and directions are processed independently, so the footprint is $O(n)$ beyond the $O(nd)$ input rather than $O(Kn)$, and scoring streams in constant-memory chunks.

b) The null gate q_0 , contamination, and heavy tails.:

q_0 enters only as a per-direction weight gate $(\Delta(u_k) - q_0)_+$ in (8): it reweights *directions* and never enters the per-point statistic τ_i , whose breakdown is carried by the median/MAD (Theorem 3), not by q_0 . We keep q_0 a Ledoit–Wolf Gaussian-copy quantile because it serves only as a relative reference, and the aggregation falls back to a uniform-weighted mean when the gate is empty. We did not run robust-covariance or permutation-based nulls; since q_0 only orders directions while robustness is decoupled into the MAD statistic, we expect low sensitivity, consistent with WAND remaining effective on heavy-tailed inliers (main paper), but a systematic robust-null study is future work.

c) High-dimensional regime and witness-cone sizes.:

The worst-case probe bound $p_\tau = \Theta((\tau/\sqrt{d})^{d-1})$ (Lemma 1) is loose in high d , yet a fixed uniform budget $K = 1024$ suffices empirically up to the highest-dimensional ADBench tasks (d up to $\approx 1.5k$), direct evidence that real witness cones are far wider than the adversarial cap. We tested adaptive (posterior-targeting) direction sampling and saw no measurable mean-AUC gain over uniform draws (Section IV-D), so we keep uniform sampling. A structure-adaptive scheme that aligns probes with the empirical principal directions is a promising way to tighten high- d behaviour and is among our stated future directions; we have not yet evaluated it.

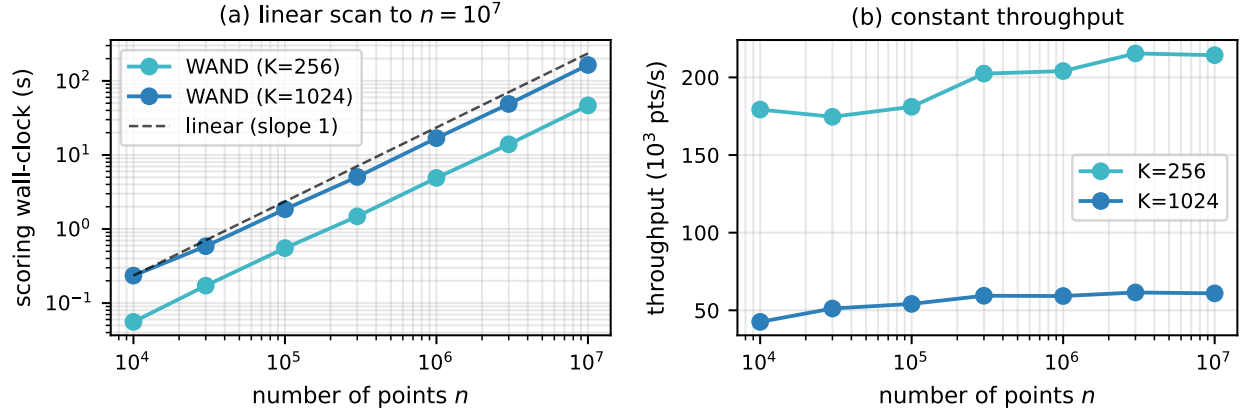


Fig. 5. Large- n scaling of the WAND scorer (synthetic, $d=20$, 8-core CPU). (a) Scoring wall-clock vs. n tracks the slope-1 (linear) reference for both probe budgets; (b) throughput is constant in n , so WAND scores 10^7 points in 47 s ($K=256$) / 164 s ($K=1024$) after a one-time ~ 1 s calibration.

d) Why the spacings penalty is small, and adaptive switching.: The MAD and spacings components are combined per point and per direction by an evidence-disjunction $\tau_i = \max(\tau^{\text{mad}}, s_u \tau^{\text{spc}})$ (6), so the spacings term contributes only where its rescaled signal exceeds the MAD term. This maximum already acts as a local, per-direction switch, which is why enabling spacings shifts mean AUC only slightly. We did not add a dataset-level gate; a cheap multimodality test (e.g. a dip statistic on $u^\top X$) that enables spacings only on multi-modal projections is a sensible heuristic to remove the residual penalty, which we leave to future work.

e) Affine transformations and the axis pathway.: Features are standardised (per-feature centring and scaling) before probing, so per-feature rescaling does not change the scores, and the uniform-sphere pathway is rotation-equivariant up to the per-direction MAD calibration. The axis pathway is, by construction, *not* affine-equivariant: it probes the canonical axes $e_1, \dots, e_d$ precisely to catch single-feature anomalies that rotation-invariant probes miss (the marginal regime), so we trade equivariance for that coverage and keep it secondary through $\lambda = 1/4$ (7). Whitening before probing is not uniformly helpful: on contaminated data it can suppress the very anomaly directions, e.g. on the ANOCUB embedding Mahalanobis whitening drops AUC from 0.98 to 0.88 (Table III); we therefore do not whiten by default.

f) Sensitivity of the faithfulness protocol.: We follow the standard deletion/insertion protocol with per-feature median replacement [6], and report the bounded, self-normalised score insertion–deletion $\in [-1, 1]$ averaged over 33 datasets, which limits the influence of any single masking choice. We did not sweep alternative imputations (mean, marginal resampling) or feature-subset sizes; such a sensitivity analysis would further strengthen the faithfulness conclusions and is straightforward future work.

g) The differentiable score as a training loss.: We use the differentiable WAND score only to produce gradient explanations, including pixel saliency through a *frozen* encoder (Fig. 1); we did not back-propagate it into the encoder as

a training objective. Using the score to shape upstream representations (a learned witness encoder) is an explicit future direction, and we report no empirical results on it here.

h) Empirical explanation coverage.: The coverage guarantee (every τ -margin anomaly has at least one witness) is realised in the attribution by the dominant-witness set. On ANOCUB all 15 anomalies are covered (detection AUC 1.0) and 14/15 concentrate on a single dominant witness (bill length), the lone exception spreading over global-shape attributes (Fig. 1). We do not yet report suite-wide distributions of the active-witness count or of the contribution mass $\gamma_{i,k}$; tabulating, per dataset, the fraction of flagged points with at least one above-threshold witness would quantify coverage directly and is a useful addition we flag for future work.

i) Parameter robustness (λ , K , k).: The defaults are insensitive over wide ranges. Mean AUC is flat (± 0.003) for $\lambda \in [0.1, 0.5]$, and we fix $\lambda = 1/4$. The budget K is a monotone speed/quality knob whose AUC saturates around $K \approx 4|H|$, with $|H|$ the soft-extreme hull size, so $K = 1024$ sits on the plateau for the suite. The spacings order $k = \lceil \sqrt{n} \rceil$ is the classical parameter-free choice [7]. A single consolidated cross-dataset robustness plot for the three knobs is not included, but the per-knob results above already indicate wide stability.

j) Comparison to recent density detectors.: Our sixteen unsupervised baselines already include strong density and marginal detectors (ECOD, COPOD, HBOS, kernel density, LODA); we did not include very recent variants such as VSDE, so we make no claim against them. Because the witness attribution needs only a differentiable score, it could in principle wrap any differentiable density model to add explanations, an interesting direction we have not evaluated.

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TABLE V

THE 47 ADBENCH TABULAR DATASETS USED IN OUR EXPERIMENTS, GROUPED BY SCALE. *Samples (n)*, *Features (d)*, *#Anom.* (NUMBER OF LABELLED OUTLIERS) AND *%Anom.* (THEIR FRACTION) ARE TAKEN FROM THE PUBLIC ADBENCH CLASSICAL MIRROR. DATASETS ARE REFERRED TO BY SHORT CODES IN THE REST OF THE PAPER; THE NUMERIC PREFIX MATCHES THE ADBENCH NAMING CONVENTION.

Dataset	<i>n</i>	<i>d</i>	#Anom.	%Anom.	Code	Dataset	<i>n</i>	<i>d</i>	#Anom.	%Anom.	Code
<i>Small-scale ($n \leq 1,000$, 12 datasets)</i>											
breastw	683	9	239	34.99%	4	glass	214	7	9	4.21%	14
Hepatitis	80	19	13	16.25%	15	Ionosphere	351	32	126	35.90%	18
Lymphography	148	18	6	4.05%	21	Pima	768	8	268	34.90%	29
Stamps	340	9	31	9.12%	37	vertebral	240	6	30	12.50%	39
WBC	223	9	10	4.48%	42	WDBC	367	30	10	2.72%	43
wine	129	13	10	7.75%	45	WPBC	198	33	47	23.74%	46
<i>Medium-scale ($1,000 < n \leq 10,000$, 15 datasets)</i>											
annthyroid	7200	6	534	7.42%	2	cardio	1831	21	176	9.61%	6
Cardiotocogr.	2114	21	466	22.04%	7	fault	1941	27	673	34.67%	12
landsat	6435	36	1333	20.71%	19	letter	1600	32	100	6.25%	20
PageBlocks	5393	10	510	9.46%	27	pendigits	6870	16	156	2.27%	28
satellite	6435	36	2036	31.64%	30	satimage-2	5803	36	71	1.22%	31
thyroid	3772	6	93	2.47%	38	vowels	1456	12	50	3.43%	40
Waveform	3443	21	100	2.90%	41	Wilt	4819	5	257	5.33%	44
yeast	1484	8	507	34.16%	47						
<i>Large-scale ($n > 10,000$, 11 datasets)</i>											
ALOI	49534	27	1508	3.04%	1	celeba	202599	39	4547	2.24%	8
cover	286048	10	2747	0.96%	10	donors	619326	10	36710	5.93%	11
fraud	284807	29	492	0.17%	13	http	567498	3	2211	0.39%	16
magic.gamma	19020	10	6688	35.16%	22	mammography	11183	6	260	2.32%	23
shuttle	49097	9	3511	7.15%	32	skin	245057	3	50859	20.75%	33
smtpt	95156	3	30	0.03%	34						
<i>High-dimensional ($d \gg 1$, 9 datasets)</i>											
backdoor	95329	196	2329	2.44%	3	campaign	41188	62	4640	11.27%	5
census	299285	500	18568	6.20%	9	InternetAds	1966	1555	368	18.72%	17
mnist	7603	100	700	9.21%	24	musk	3062	166	97	3.17%	25
optdigits	5216	64	150	2.88%	26	SpamBase	4207	57	1679	39.91%	35
speech	3686	400	61	1.65%	36						

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